Realtime Motion Capture as an Automated MoBI Quality Control Mechanism

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Challenge

Children with **neurodevelopmental disorders**, including ADHD and ASD, often exhibit **increased motor activity**, **fidgeting**, and **wandering head movements**, complicating eye tracking, physiological signals, and EEG data[1,2].

Is degraded data due to technical failure, movement-related artifacts, or meaningful within-session variability?

The ZED 2i Motion Classifier

Open-Source Python Pipeline: Markerless 3D body tracking via Stereolabs ZED 2i camera.

No Wearables Required: Operates continuously at 30 Hz without restricting participant movement.

Multimodal Sync: Streams per-frame & windowed labels via Lab Streaming Layer (LSL).

Core Outputs

1. Fidgeting Intensity Score

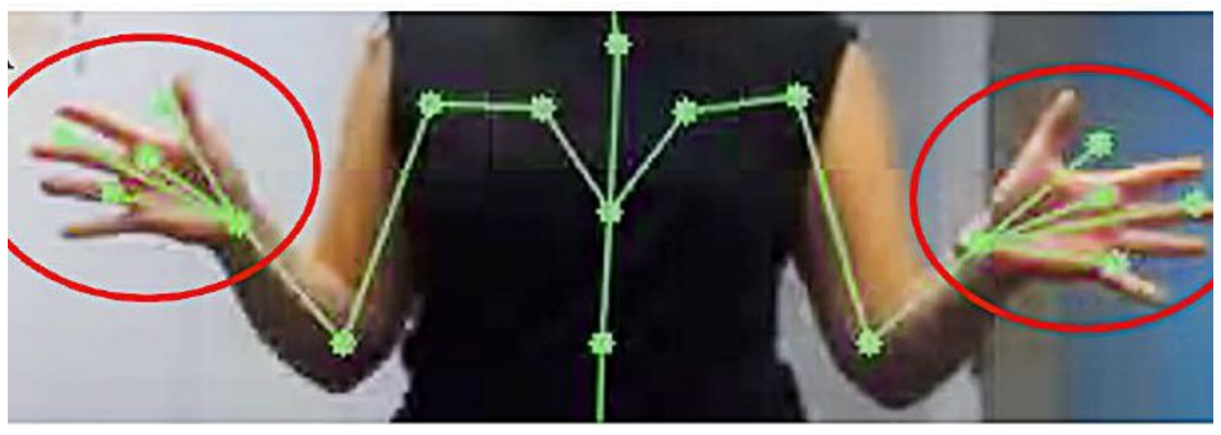

Figure 1: Upper-body skeletal tracking with detailed hand joint estimation, demonstrating capture of fine-grained limb and finger movements used for classifying behavior (fidgeting).

2. Postural Shift Detection

Figure 2: Skeletal tracking with joint estimation, demonstrating capture of body movements used for classifying postural shifts (leaning forward and to the sides).

3. Head Direction Vectors

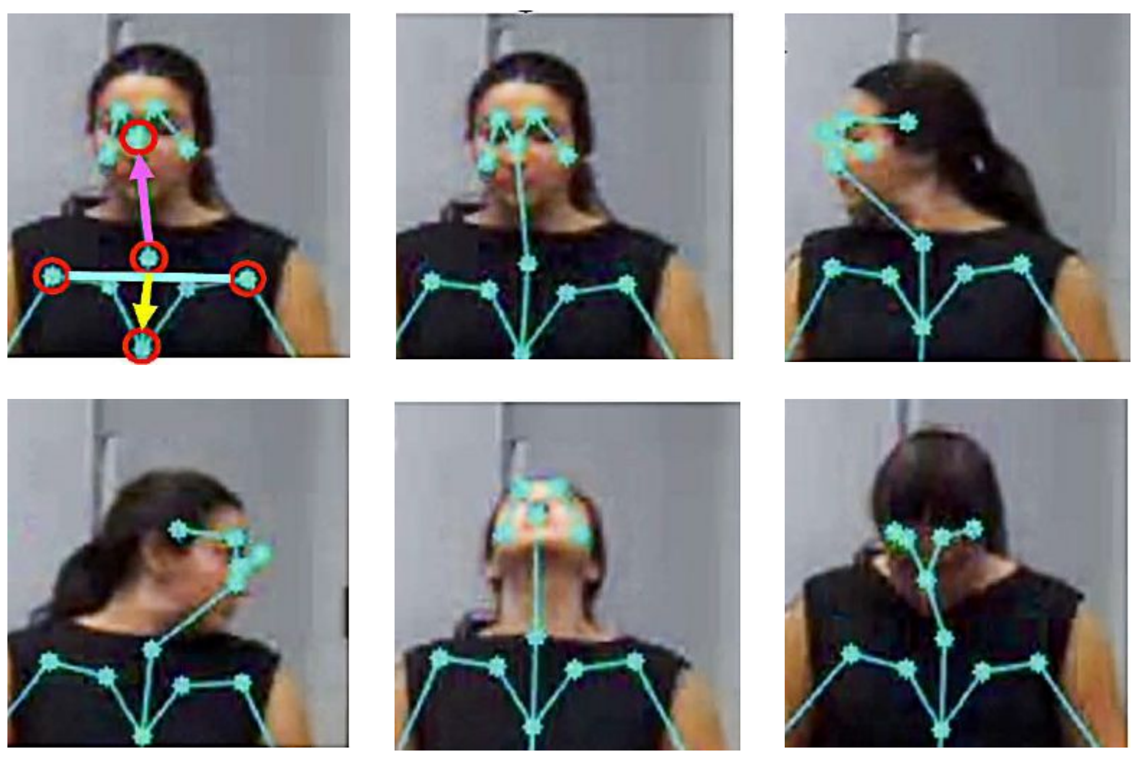

Figure 3: Schematic illustrating the vectors derived from facial and torso landmarks used to compute head orientation. Head and upper body orientation changes for forward, left, right, up, and down positions.

Advantages

LSL Stream
Synchronized with all data streams

Real-Time Labels
Behavior, Posture, and Head

30 Hz Stereo
Markerless 3D Tracking

Open-Source
Code and documentation

Automated Quality Control:
Real-time movement artifact identification and trial rejection.

Natural Behavior:
Preserves unconstrained movement in pediatric testing.

Phenotyping Marker:
Motion features can serve as markers of postural and attentional control, capturing meaningful behavioral phenotypes when combined with neurocognitive measures[3].

Easy to set up
Uses uv package manager

Take home message

Continuous and real-time non-invasive measures of movement intensity, postural stability, and head orientation for synchronization with multimodal physiological recordings.

SCAN ME !

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References:
[1] Thapar et al., 2015.
[2] Church et al., 2010.
[3] Wilson et al., 2024.